GraphIV: A Grounded, Citation-Graph-Native Literature-Review Engine over arXiv

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Abstract

We present **GraphIV** (graph + ar*Xiv*), a system that builds a knowledge graph over the entire arXiv corpus — a 173M-edge citation graph with classified citation *intent*, 3.07M dense embeddings, and a density-aware topic tree — and serves it as a research endpoint that turns a natural-language question into a grounded, cited, chronological literature review. Unlike generic retrieval-augmented generation, GraphIV is *citation-graph-native*: it induces a bounded on-topic subgraph, ranks landmark papers by Personalized PageRank with topic-adaptive recency, and composes a review whose every claim is tied to an arXiv id. We back every design choice with a measured experiment suite (preprocessing, retrieval, generation), and report a recurring failure mode of citation-graph retrieval — *generic-hub dominance*, where globally-ubiquitous papers (e.g. "Attention Is All You Need", cited in 27/27 of our deep reports) hijack niche reviews — which we show is not cleanly solvable in retrieval and address with a validated compose-side guard. End to end, the full pipeline is preferred over a strong single-pass baseline **8/0/0** (win/tie/loss) on overall and on grounding by an order-debiased pairwise judge reading the full reports, a verdict confirmed unanimously by two disjoint-family judges (deep panel, n=8; sign test $p \approx 0.008$ per judge).

1 Introduction

Keeping up with a fast-moving field means reading its literature as a *development* — origins, key advances, current frontier — not as a ranked list. Large language models can write fluent surveys but hallucinate citations and lack the structural signal that a citation graph carries. We ask: can a system over the *whole* arXiv corpus produce a literature review that is (i) grounded in real papers with verifiable citations, (ii) chronologically and causally organized via the citation graph, and (iii) specific to a niche query rather than padded with generic background?

GraphIV answers this with a six-phase pipeline on a single Postgres+pgvector store. Phases 1–5 acquire PDFs/HTML, parse and intent-classify the corpus-wide citation graph, consolidate metadata + 3.07M nomic embeddings, and build a density-aware topic tree. Phase 6 is the live endpoint. Our contributions: (1) a citation-graph-native retrieve–rank–compose pipeline that grounds every claim; (2) a measured experiment suite that justifies each lever as a metric *vector* rather than a single gameable number; and (3) a diagnosis and validated fix of *generic-hub dominance*.

2 System

The store. One project-local Postgres+pgvector database holds `paper` (id↔ arxiv_id), `citation` (173M edges with intent/polarity), `paper_meta`, `paper_embedding` (3.07M × 768-d nomic vectors, HNSW), and `paper_topic/leaf` (3,582 leaves over 2.92M papers). A DistilBERT student (distilled from a Gemma teacher over 100k pseudo-labels) classifies every citation's intent (background / uses / extends / compares / critiques / motivates) and polarity.

The endpoint. A question is embedded with an asymmetric query/document encoder; retrieval fuses a vector pool, lexical signals, a topic-tree leaf anchor, an induced citation subgraph, and a leaf-independent recent-paper pull (the topic tree predates the corpus delta). Landmarks are ranked by Personalized PageRank seeded by query similarity, with a topic-adaptive frontier reservation. Per-paper schema fields are extracted from full text, and a Gemma-Code model composes a chronological, grounded review; an `effort` dial (0–5) trades latency for a judge-driven deep loop with full-PDF critique and a caption-faithfulness-checked figure gallery. Invalid arXiv ids are stripped post-hoc.

3 Evaluation methodology

We evaluate over a 27-topic battery (broad fields + niche compositional probes) with three gold topics (diffusion, transformers, test-time compute) carrying curated canonical/spine sets. Every experiment reports a metric *vector*: retrieval recall/precision, citation grounding, an independent 4-axis LLM judge (grounding/coverage/lineage/coherence) plus an order-debiased pairwise, a faithfulness tripwire, and caption-faithfulness. We validate the judge itself (known-fact accuracy, no position/verbosity bias; lineage is its weak axis) before trusting it downstream.

4 Key results

Retrieval. The committed subgraph recipe recovers 0.74 of canonical landmarks and 0.92 of the foundational spine against a 0.85 pool ceiling. A role judge finds ∼78% of landmarks on-topic; the ∼22% off-topic remainder are famous cross-field hubs that carry high in-degree. A cosine floor on the most-cited-ancestor injection is a Pareto win (precision and recall both up): a 0.25–0.45 sensitivity sweep shows recall flat across 0.25–0.35 with precision rising monotonically, so the shipped default (0.35) is the max-precision edge of a plateau, not a tuned peak; the cliff starts at 0.40.

Generation. Component ablation confirms no single metric can be trusted: dropping the citation graph *maximizes* a surface "frontier" count while zeroing edges and recall and producing the worst judge score. An asymmetric encoder without HyDE wins on recall at no extra cost. The judge-driven deep loop is a real quality lever: pairwise 0.75 over single-pass, with the thinking judge self-terminating. Full-PDF critique depth beyond a few is a documented null.

Generic-hub dominance (Section 5). The flagship's dominant rough edge.

Flagship. The full deep pipeline vs. a strong single-pass baseline: preferred **8/0/0** on overall and on grounding when the pairwise judge reads the *full* reports, confirmed unanimously by two disjoint-family judges (zero baseline wins across 96 axis-verdicts); it surfaces ∼9 2026 papers/topic where the baseline surfaces zero, at ∼5× the length with verified figures. Notably, an earlier 16k-character judge truncation — which saw only half of each deep report — yielded a much softer 5/2/1: a measurement-instrument artifact worth reporting in its own right.

5 Generic-hub dominance

Reading real outputs, "Attention Is All You Need" appears in **27 of 27** deep reports, BERT in 17, GPT-3 in 16, the Adam optimizer in 11 — regardless of topic. A theorem-proving review devotes a section to Attention→BERT→GPT-3 and cites Adam as "foundational to theorem proving". Cause: these hubs accrue high *in-topic* in-degree, so PPR floats them to top landmarks. This is a *specificity* problem, and we require a fix with no hand-curated blocklist.

We swept every data-derived demotion signal (global IDF, topic-lift $P(\text{cited} \mid \text{topic})/P(\text{cited})$, citation-intent background-ratio, a personalized-PR/global-popularity penalty, and a topic-conditioned

"demote iff popular AND background-cited-here" hybrid), scored against gold-recall (tripwire) and an independent topic-conditioned LLM oracle. The result is a genuine recall↔specificity *frontier*: the topic-conditioned penalty holds recall perfectly but is gentle; topic-blind penalties move the oracle most but drop papers that are canonical for hub-y topics. **No purely-structural signal removes a hub that is genuinely *used* in a niche** — theorem-proving papers really do use transformers — so "generic for this topic" is partly a *narrative* judgment. The shipped fix is therefore a one-line compose guard forbidding a generic-foundations section unless directly load-bearing. A deep A/B removes the boilerplate section and drops Attention/BERT/GPT-3/Adam from 3/1/3/1 to **0/0/0/0** mentions while preserving the topic-specific coverage; validated on three further niche topics.

6 Limitations

Curated gold exists for only three topics (recall numbers are n=3); the LLM judge, though validated, compresses on its absolute scale (we lean on order-debiased pairwise and objective metrics), and its primary instance shares a model family with the generator — mitigated for the flagship result by the unanimous disjoint-family confirmation. The topic tree predates the most recent corpus delta (mitigated by the frontier pull). The hub fix is narrative, not structural; a recall-safe retrieval lever for the milder generic-*method* residue is available but not enabled by default.

7 Conclusion

GraphIV shows that a citation-graph-native, fully-measured pipeline can turn a question into a grounded, specific, chronological literature review over all of arXiv, and that the dominant quality failure of such systems — generic-hub dominance — is best understood and fixed at the composition stage. The system, the experiment suite, and the open resource are released under the GNU GPLv3.